

Tanurima Halder

+91 9475995498 | haldertanurima98@gmail.com | linkedin.com/in/tanurima | github.com/blackhorse754 | Hyderabad, India

Summary

AI/ML Engineer with M.Tech in Artificial Intelligence from IIT Hyderabad specializing in deep learning, NLP, and generative AI systems. Experienced in building production-grade RAG pipelines, transformer-based NLP models, and TinyML edge systems. Published researcher at IEEE ICKG, ISBI, and ACL SemEval with work in federated learning and privacy-preserving machine learning.

Technical Skills

Programming: Python, C++, SQL, MATLAB

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Model Optimization

LLM & Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, OpenAI API, HuggingFace Transformers

NLP: Transformer Models, Text Classification, Semantic Search, Lexical Complexity

Data & Tools: NumPy, Pandas, Docker, Git, Linux, SQLite

Edge AI / TinyML: MobileNetV1, Once-for-All Networks, Model Compression

Experience

Data Scientist

George Institute for Global Health

July 2025 – April 2026

Hyderabad, India

- Developed production multi-turn RAG chatbot using LangChain and OpenAI GPT to assist ASHA workers with pregnancy-related clinical queries.
- Implemented retrieval pipelines grounding responses in Indian government medical guidelines.
- Built analytics pipeline tracking session duration, query classification accuracy, and user retention metrics.

Pre-Doctoral Researcher

IIT Bombay

July 2024 – July 2025

Mumbai, India

- Conducted research on federated learning and privacy-preserving machine learning for medical imaging datasets.
- Developed MRI harmonization models using disentangled representation learning.

Projects

Multi-Turn Conversational Medical RAG Chatbot

Python, LangChain, OpenAI GPT, SQLite

2025

- Designed retrieval-augmented conversational pipeline for healthcare decision support.
- Integrated semantic retrieval and LLM reasoning to provide context-aware responses.
- Built analytics framework monitoring error rates, engagement metrics, and system performance.

Radar-Driven TinyML Gesture Detection System

Python, PyTorch, MobileNetV1, Once-for-All

2024

- Developed radar-based gesture recognition model optimized for edge devices.
- Achieved 90% classification accuracy while reducing power consumption by 40%.

Publications

Halder et al. **When to Harmonize? Evaluating Stage-Specific Harmonization in Federated Brain Age Estimation.** IEEE ICKG 2025.

Galada, Halder et al. **PRISM: Privacy-Preserving Inter-Site MRI Harmonization via Disentangled Representation Learning.** ISBI 2025.

Nandy, Halder et al. **Lexical Complexity Prediction using Transformer-based Language Models.** ACL SemEval 2021.

Education

IIT Hyderabad

M.Tech in Artificial Intelligence and Machine Learning

Hyderabad, India

Aug 2022 – Jul 2024

Jalpaiguri Government Engineering College

B.Tech in Computer Science and Engineering

Jalpaiguri, India

Aug 2016 – Jul 2020